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Short Communication

The AI challenge: A turing test pilot study of attendings and residents in identifying AI-Generated content

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ABSTRACT

Generative Artificial Intelligence (AI) models have demonstrated strong potential in radiology report generation, but their clinical adoption depends on physician trust. In this pilot study, we conducted a radiology-focused Turing test to evaluate how well attendings and residents distinguish AI-generated reports from those written by radiologists, and how their confidence and decision time reflect trust. We developed an integrated web-based platform for report evaluation. Using the web-based platform, eight participants (4 attendings and 4 residents) evaluated 48 anonymized X-ray cases, each paired with two reports from three comparison groups: radiologist vs. AI model 1, radiologist vs. AI model 2, and AI model 1 vs. AI model 2. Participants were asked to select the AI-generated report, rate their confidence, and indicate report preference. Results show that attendings outperformed residents in identifying AI-generated reports (49.9 % vs. 41.1 %) and exhibited longer decision times, suggesting more deliberate judgment. Both groups took more time when both reports were AI-generated. Our findings highlight the role of clinical experience in AI acceptance and the need for design strategies that foster trust in clinical applications.

1. Introduction

In recent years, rapid advances in generative AI, supported by the evolution of large vision models and visual prompt engineering,[1] have accelerated progress in medicine, particularly in radiology, where these systems show strong potential for automating the labor-intensive task of report generation. Generative AI can produce radiology reports that are fluent, clinically coherent, and in many cases comparable to or even preferred over human-written reports, highlighting their clinical utility.[2,3] Moreover, the integration of multimodal large language models with both imaging and textual data[4] has shown promise in reducing reading time and improving report accuracy and consistency, thereby enhancing the efficiency and quality of chest X-ray interpretation.[5]

Despite their promise, the integration of generative AI into clinical workflows depends critically on physicians' acceptance and trust. Recent studies reveal a concerning nuance: both experts and non-experts sometimes perceive AI-generated medical content as more thorough and accurate than physician-written responses, even rating low-accuracy outputs as valid.[6] Such misplaced trust heightens the risk of misdiagnosis or unnecessary interventions when AI errors go undetected. These findings underscore the dual challenge of fostering appropriate trust: encouraging physicians to adopt AI when it is reliable while preventing blind reliance on inaccurate outputs. Importantly, physicians' level of clinical experience may shape how they perceive and evaluate AI reports, raising a critical but underexplored question: to what extent does clinical experience influence the ability to identify,

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2950-1628/© 2025 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

trust, or challenge AI-generated content? Addressing this question is essential for guiding the safe adoption of AI in medical environment.

To explore these issues, we designed a radiology-oriented Turing Test as a rigorous experimental framework. The classical Turing Test was originally conceived to evaluate whether machines exhibit intelligence by assessing if humans can distinguish machine-generated outputs from those produced by humans. Following this principle, our study asks physicians to determine whether radiology reports are authored by radiologists or generated by AI systems. By directly testing the distinguishability of AI- versus radiologist-produced reports, our framework adapts the Turing Test to the clinical domain. Beyond classification accuracy, our experiment allows for a deeper analysis of decision-making processes, such as judgment time and confidence, which can promote further research on the cognitive mechanisms underlying physicians' trust in AI. By comparing attending physicians with residents, our study also provides insight on how clinical experience shapes attitudes toward AI, hopefully providing insights into fostering effective physician–AI collaboration in medical environments.

2. Materials and methods

We developed a web-based evaluation platform to assess physicians' trust and acceptance of AI-generated radiology reports using a Turing-test-inspired design.[7] As shown in Fig. 1, the evaluation pipeline integrates radiologists, AI models, and an evaluation interface to systematically compare human- and AI-generated reports. The study included 48 chest X-ray cases, each with a corresponding radiologist report. For each case, two additional reports were generated using state-of-the-art methods, MRANet[8] and SEI[9]. To ensure a fair and unbiased evaluation, we implemented multiple quality control measures during the generation and presentation of radiology reports. First, all reports were presented in a standardized plain-text format, with formatting artifacts such as timestamps, or headers removed. Second, both models were pretrained on the MIMIC-CXR[10] dataset (269,239 chest X-rays with paired free-text reports) and subsequently fine-tuned on 1000 chest X-rays from Rhode Island Hospital to match the local reporting style and structure. Lastly, we selected state-of-the-art models that have demonstrated strong performance in radiology report generation tasks, thereby reducing the risk of obvious AI artifacts common in earlier or less domain-adapted systems. Importantly, no manual editing was performed to remove AI artifacts in generated outputs, such as language redundancy or factual inconsistencies, as these features are critical to physicians' decision-making in the Turing test setting.

These 48 cases were randomly divided into three experimental conditions: (1) radiologist vs. AI model 1, (2) radiologist vs. AI model 2, and (3) AI model 1 vs. AI model 2, with 16 cases per condition. In each case, a medical image (e.g., chest X-ray or CT slice) was displayed at the top, accompanied by two side-by-side diagnostic reports labeled *Report 1* and *Report 2*. Report pairs were constructed according to the assigned condition: in one-third of cases, a radiologist report was paired with AI model 1, in another third a radiologist report was paired with AI model 2, and in the final third both reports were AI-generated. The assignment of a report to the left (*Report 1*) or right (*Report 2*) position was randomized to prevent positional bias.

Participants were asked to review the image and reports, then answer three questions: (1) Which report do you believe was generated by AI? Participants were given three response options: "Report 1 is generated by AI," "Report 2 is generated by AI," or "Both reports are generated by AI." (2) How confident are you in your judgment? (3) Which report would you prefer to use in clinical practice? To ensure fair comparison, report formatting was standardized, the identities of reports (AI vs. human) were concealed, and no model or author identifiers were revealed.

Eight radiologists (4 attendings and 4 residents) participated in the study. The attendings had between 8 and 29 years of post-residency clinical experience (mean: 17.5 years) and subspecialized in areas such as Cardiac Radiology, Chest Radiology, Body Imaging, and Nuclear Medicine. The residents were in their first to third year of radiology training (mean clinical experience: 2.5 years) and had not yet declared formal subspecialties. None of the participants had prior experience using AI-assisted report generation tools in clinical settings. For each case, participants indicated which report they believed was AI-generated, rated their confidence on a 5-point Likert scale (1 = low, 5 = high), and selected which report they would prefer for clinical use. Responses were analyzed using generalized estimating equations. The study was approved by the institutional review board (IRB #2055965-7).

3. Results

3.1. Performance in identifying AI-generated reports

Fig. 2 shows the mean accuracy of attendings and residents in identifying AI-generated reports, with error bars representing 95 % confidence intervals. Attendings achieved an average accuracy of 49.0 %, whereas residents averaged 41.2 % ($p = 0.23$). Both rates exceeded the random chance level of 33.3 % (i.e., 1 in 3 chance of guessing correctly), which reflects the three-choice design of the task (Report 1, Report 2, or Both reports), indicating that radiologists were generally able to surpass the Turing test, with attendings performing better than residents. This finding suggests that greater clinical experience enhances the ability to discriminate between AI- and radiologist-generated reports, whereas residents may be more susceptible to perceptual biases.

To further analyze performance across different task conditions, we stratified identification accuracy by comparison group: Radiologist vs. AI 1(SEI), Radiologist vs. AI 2(MRANet), and AI 1 vs. AI 2. As shown in Fig. 2(b), both attendings and residents achieved higher accuracy when the comparison involved a human-written report. Specifically, in the Radiologist vs. AI 1 condition, attendings and residents achieved 45.3 %

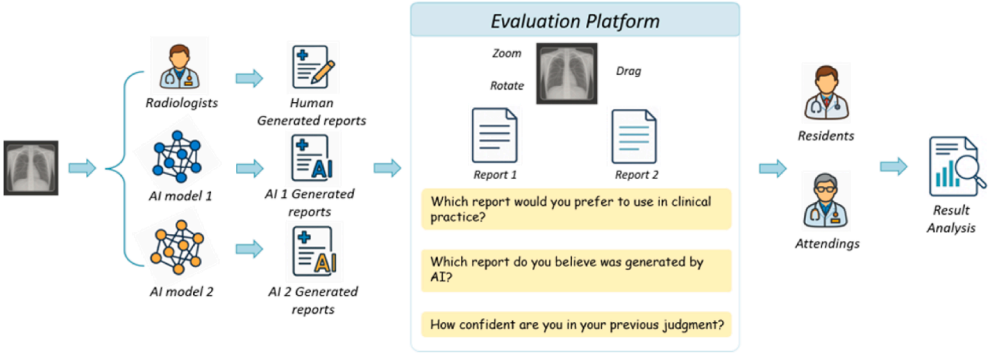


Fig. 1. Overview of the evaluation pipeline for AI-Generated reports turing test.

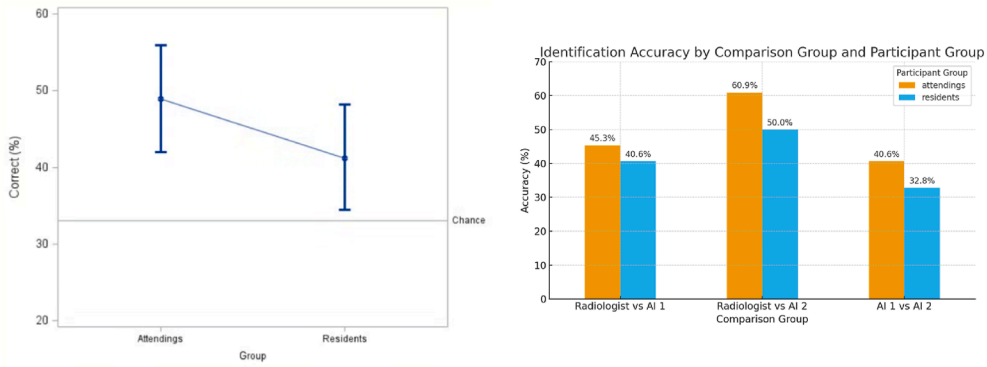


Fig. 2. Accuracy of attendings and residents in identifying AI-generated reports. (a) Mean correct response rates are shown with 95 % confidence intervals. Attendings achieved higher accuracy (≈49 %) than residents (≈41 %), and both groups performed above the chance level (33.3 %, horizontal line). (b) Stratified identification accuracy by comparison group: Radiologist vs. AI 1 (SEI), Radiologist vs. AI 2 (MRANet), and AI 1 vs. AI 2. Attendings (orange) consistently outperformed residents (blue) across all conditions. Reports generated by SEI were more difficult to distinguish from human-written ones, as reflected in lower accuracy in the Radiologist vs. AI 1 group. In contrast, MRNet-generated reports were more easily identified as synthetic, leading to higher accuracy in the Radiologist vs. AI 2 group.

and 40.6 % accuracy, respectively. Performance improved in the Radiologist vs. AI 2 condition (60.9 % for attendings and 50.0 % for residents), suggesting increased distinguishability between the radiologist and the second model(MRANet). In contrast, when both reports were AI-generated (AI 1 vs. AI 2), accuracy dropped considerably, particularly among residents (32.8 % vs. 40.6 % for attendings), and approached the theoretical chance level of 33.3 %. The fact that participants struggled to differentiate between AI 1 and AI 2 suggests that AI-generated reports, despite being produced by distinct models with different architectures and generation mechanisms, may converge toward similar stylistic patterns, clinical templates, or linguistic fluency that mask their underlying differences.

As shown in Table 1, attendings also exhibited significantly longer average response times than residents (56.8 vs. 31.9 s, $p = 0.0034$), consistent with a more deliberate decision-making approach. Fig. 3 illustrates that response times increased further in the AI–AI comparison condition, rising by 18.9 % for attendings and 28.9 % for residents, suggesting that fully synthetic case pairs imposed greater cognitive load and diagnostic uncertainty.

Confidence ratings were marginally higher for attendings than for residents, although the difference was not statistically significant ($p = 0.4830$). Across both groups, each one-unit increase in confidence was associated with a 22 % higher odds of correctly identifying AI-generated reports (Odds Ration (OR) = 1.22, 95 % Confidence Interval (CI) [1.01–1.50], $p = 0.0386$).

3.2. Error analysis and trust bias

To further interpret the evaluation outcomes, we conducted an error-type analysis to examine how physicians misclassified AI-generated and human-written reports. Two primary types of misjudgment were identified: (1) *false acceptance of AI* (both reports were AI-generated, but participants selected "1 is AI-generated" or "2 is AI-generated.") and (2) *false rejection of AI* (the ground truth included one AI-generated report and one human-written report, but participants selected "both are AI-generated."). Together, these patterns illustrate two opposing forms of trust bias: overtrust, in which physicians incorrectly accept AI reports as genuine, and over-skepticism, in which they fail to recognize authentic human work.

Building on this classification, we quantified the distribution of these errors across attendings and residents, as shown in Table 2. Residents showed 31 false acceptances of AI and 29 false rejections, whereas attendings made 26 and 24 such errors, respectively.

Interestingly, the confidence ratings associated with these errors reveal subtle cognitive asymmetries: both groups exhibited higher confidence when falsely accepting AI-generated reports (3.87 for residents, 3.69 for attendings) than when falsely rejecting human-written ones (3.20 and 3.21, respectively). This suggests a trust bias toward algorithmic outputs, probably driven by the polished linguistic fluency and structural coherence of AI-generated text. Such stylistic coherence may inadvertently enhance the perceived credibility of AI-generated content, even when the underlying factual accuracy is compromised. This pattern was observed across both experience levels, suggesting that overconfidence in AI-generated reports is not confined to less experienced clinicians.

These findings illustrate the deceptive realism of AI-generated radiology reports, which can evoke misplaced confidence even among experienced physicians. Conversely, the tendency to misclassify authentic reports as AI-generated indicates a growing skepticism toward algorithmic outputs. Together, these results emphasize the importance of developing AI systems that promote calibrated trust—enabling physicians to maintain an optimal balance between appropriate reliance on AI and critical clinical judgment.

Table 1
Comparison of accuracy, response time, and confidence between residents and attendings in identifying AI-Generated reports.

Participants	Accuracy	Response Time	Confidence Level
Attendings	49.0 %[44.8 %, 53.2 %]	56.8[43.5, 74.3]	3.8[3.4, 4.2]
Residents	41.2 %[28.8 %, 54.8 %]	31.9[27.7, 36.6]	3.6[3.3, 3.9]

4. Discussion

4.1. Performance and decision-making differences

Our findings provide several insights into how physicians interact with AI-generated radiology reports. It is important to note that our study design differed from conventional binary Turing-test settings. Because one-third of the cases involved AI–AI comparisons, participants had three possible response options, making the chance level 33.3 % rather than 50 %—a more rigorous design. The fact that both attendings and residents performed above chance suggests that current AI models have not yet achieved full indistinguishability in clinical reporting. Although attendings achieved higher accuracy than residents, highlighting the possible role of clinical experience in detecting subtle linguistic or structural cues that may differentiate AI outputs from radiologist reports, this difference failed to be significant.

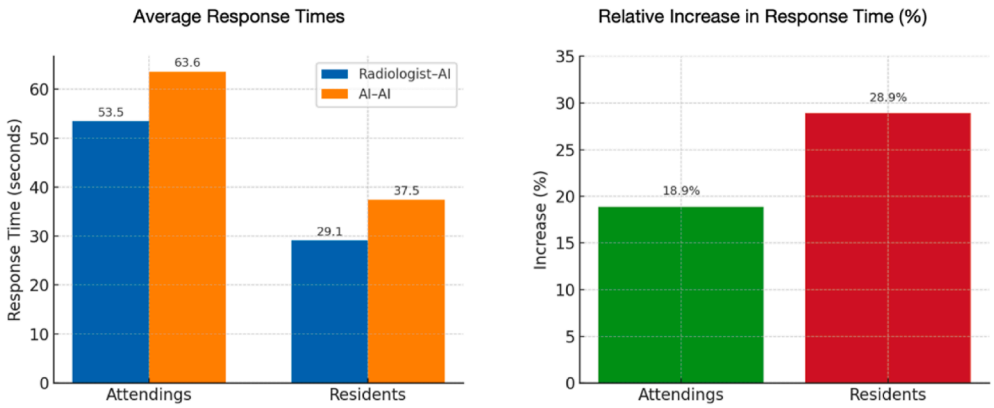


Fig. 3. Average response times of residents and attendings across different task conditions. The left panel shows average response times under Radiologist–AI and AI–AI conditions. The right panel shows the relative increase in response times for AI–AI compared with Radiologist–AI.

Table 2
Comparison of error types and confidence ratings between residents and attending physicians.

Participant	Number of Errors		Average Confidence	
	False Acceptance of AI	False Rejection of AI	False Acceptance of AI	False Rejection of AI
Residents	31	29	3.87	3.20
Attendings	26	24	3.69	3.21

The observation that attendings required significantly longer response times than residents suggests that they adopted a more deliberate and analytical decision-making approach. In contrast, residents appeared to rely on quicker, potentially more impression-driven judgments. This tendency may help explain why residents were more susceptible to perception-driven preferences. For example, they may have placed greater weight on surface-level linguistic fluency or stylistic cues rather than on deeper clinical content. While our study highlights this difference, the underlying mechanisms driving residents' susceptibility were not fully explored and warrant further investigation.

Notably, although attendings spent more time on average than residents (56.8s vs. 31.9s), no significant correlation was found between response time and identification accuracy in either group. To formally evaluate this relationship, we conducted Pearson correlation analyses, which yielded $r = -0.04$ ($p = 0.49$) for attendings and $r = 0.09$ ($p = 0.23$) for residents. These results suggest that longer evaluation time did not systematically improve discrimination performance. Therefore, it remains uncertain whether extended evaluation actually enhances judgment or simply reflects increased cognitive effort. Future studies with larger samples and more granular behavioral metrics will be needed to disentangle the relationship between time spent and diagnostic accuracy.

Beyond response time and experience level, differences in model performance also impacted participants' ability to distinguish AI-generated from human-written reports. Specifically, participants found it more difficult to distinguish reports generated by SEI (AI 1) from those written by radiologists, as reflected in lower identification accuracy (45.3 % for attendings and 40.6 % for residents). In contrast, performance improved in the Radiologist vs. AI 2 condition (60.9 % for attendings and 50.0 % for residents), suggesting that MRANet (AI 2) produced outputs that were easier to identify as AI-generated. These results imply that SEI's design, which includes structured clinical entity extraction and stylistic noise reduction, may have enabled it to produce more natural and clinically coherent reports. Consequently, participants may have primarily relied on linguistic structure and clinical expression when making their judgments. These mechanisms may have enhanced the naturalness and trustworthiness of the reports, thereby blurring the perceptual boundary between human and AI-generated content.

4.2. *Identification criteria and confounding factors*

In keeping with the original setup of the Turing Test, we intentionally did not provide participants with explicit guidelines on how to distinguish AI-generated from human-written reports. This design allowed participants to rely on their own judgment, consistent with the Turing Test's aim to evaluate indistinguishability without predefined criteria.

However, because this was an initial pilot study, our survey did not include targeted questions to capture the specific features or heuristics participants relied on during their evaluations. As a result, we were unable to systematically analyze or disentangle potential confounding factors, such as reliance on linguistic fluency, clinical accuracy, or stylistic cues, which may have influenced participants' judgments in different ways.

We acknowledge this as a key limitation of the current study and plan to address it in future work by incorporating targeted survey questions and post-task interviews. This will help us better understand the specific reasoning strategies clinicians rely on when determining which report was AI-generated and clarify how different heuristics contribute to successful or failed identification.

4.3. *Confidence and judgment reliability*

Finally, we examined whether physicians' self-perceived confidence reflected the reliability of their judgments. A logistic regression analysis revealed that each one-unit increase in confidence was associated with a 22 % higher odds of correctly identifying AI-generated reports (Odds Ratio = 1.22, 95 % CI [1.01–1.50], $p = 0.0386$). This statistically significant association suggests that confidence ratings can, to some extent, reflect judgment accuracy. When physicians feel more certain, their decisions are generally more trustworthy. This finding suggests that in future AI-assisted diagnostic workflows, confidence ratings could serve as a useful indicator: when confidence is low, additional human oversight or second readings should be considered.

5. Conclusion

In this pilot study, we developed a Turing-test-inspired web platform to evaluate physicians' ability to distinguish AI-generated from human-authored radiology reports. Both attendings and residents performed above chance, with attendings achieving higher accuracy and adopting more deliberate decision-making strategies. Physicians' self-reported confidence was positively correlated with accuracy, suggesting that

confidence may serve as a practical marker of diagnostic reliability and could be incorporated into future AI–clinician workflows.

Despite its limitations, this platform offers an early framework for understanding how clinical experience, decision-making processes, and confidence shape physician interaction with AI-generated reports. Our findings highlight important possible differences between attendings and residents in acceptance and bias toward AI outputs, underscoring the need for targeted strategies to foster clinician trust and ensure the safe and effective integration of AI tools into clinical practice. Future research should extend validation to larger and more diverse physician populations, additional imaging modalities, and a broader range of AI systems.

CRediT authorship contribution statement

Zhuoqi Ma: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Xinye Yang:** Software. **Zachary Atalay:** Software. **Zhusi Zhong:** Data curation. **Scott Collins:** Data curation. **Harrison Bai:** Formal analysis. **Michael Bernstein:** Formal analysis. **Ankur Pandey:** Investigation. **Eric Dietsche:** Investigation. **Joey Gu:** Investigation. **Jothika Challapalli:** Investigation. **Michael Atalay:** Investigation. **Mohammad Abubaker-Sharif:** Investigation. **Terrance Healey:** Investigation. **Venkata Paruchuri:** Investigation. **Grayson Baird:** Writing – review & editing, Supervision, Conceptualization. **Zhicheng Jiao:** Writing – review & editing, Supervision, Conceptualization.

Ethic statement

Institutional Review Boards at RIH (Rhode Island Hospital) approved the study, and informed consent was waived due to the retrospective nature of the study.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The author Zhicheng Jiao is the Editorial Board Member of the journal, but was not involved in the peer review procedure. This paper was handled by another Editor Board member.

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List of abbreviations

AI: Artificial Intelligence
MRANet:: Multi-Modality Regional Alignment Network for Covid X-Ray Survival Prediction and Report Generation
SEI: Structural Entities Extraction and Patient Indications Incorporation for Chest X-Ray Report Generation
MIMIC-CXR: MIMIC Chest X-ray
OR: Odds Ration
CI: Confidence Interval